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Semester Project – Case Study Report

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I. Introduction

Caribe Surgical Access, a medical-device manufacturer with facilities in Mayagüez, Puerto Rico, hired this engineering team to evaluate incorporating T12 into the existing facility. Currently, the facility produces two products, T5 and T10. Similar to T5 and T10, T12 is a surgical access device used to create and maintain a small opening through the abdominal wall during minimally invasive surgery (CSA). Additionally, to this team's advantage, T5, T12, and T10 share a cannula and obturator assembly and packaging module. Although they share the same assembly, the design team must introduce the new line within established constraints, as discussed further in this report. The purpose of this report is to support the seamless introduction of T12 and increase production of the product family through years one to five.

II. Project and facility-design problem definition

In Year 1, CSA has to go from 192,000 to 266,400 good devices per year, which is about 39% more, while also adding a third product, T12. All of this must happen in the same building and on a single shift (CSA, Table 1). By Year 5, demand goes up to 325,000 devices, or about 69% more than today. So, the design problem is figuring out what must change in the existing facility so it can handle this growth and the extra product without breaking any of its fixed physical constraints.

Design Need

- Right now, the Mayagüez plant makes two single-use trocars, T5 and T10, in a 220 ft by 120 ft building (26,400 ft²). The plant molds its own plastic components, machines the stainless-steel shafts, buys seals and packaging materials, and then assembles, tests and packages each device before sending it to an outside sterilizer (CSA, Section 1 and Section 2d).
- This matters because T12 is not just more of the same. It has its own cannula, shaft, and label, and its process times are longer than those of T5 and T10 (CSA, Tables 2 and 6–8). If the facility cannot take on this volume and mix, CSA will not be able to meet Year-1 demand under normal operating conditions.
- What the company needs is a facility that meets Year-1 demand on one shift and also shows a realistic path, both physical and in capacity, toward the Year-5 mix. The case

also asks that the existing building be used first before looking at expansion (CSA, Section 3).

Objectives

1. Meet Year-1 demand of 266,400 good units per year working one normal shift (420 net minutes per day).
2. Bring in T12 while taking advantage of the architecture all three products share (cannula, obturator and packaging) and of the parts T10 and T12 already have in common.
3. Improve how material moves from receiving on the west side to shipping on the east side, with fewer aisle crossings and less backtracking.
4. Protect the conditions inside the controlled assembly area and keep product separated by size and by status (good, rework, scrap and quarantine).
5. Leave room to grow toward Year 5, first inside the current building and then, only if it is justified, into the 8,800 ft² reserve on the north side.

System Boundary and Level of Study

	In scope	Out of scope
Physical	Existing 26,400 ft² building and the 220 × 40 ft north expansion reserve as a future option	Building construction and engineering
Process	From the receiving docks (west) to the shipping docks (east): storage, molding, machining, inspection, kitting, assembly, test, rework, packaging, and finished goods	Sterilization (external contractor), distribution to the customer
Product	T5, T10, T12 family and its MAKE/BUY structure	Clinical performance, regulatory submissions, mold and tool design
Support	Services the production system needs: quality, maintenance, utilities, offices, personnel	Sterilization technology details

Table 1. System boundary and level of study

We are looking at the facility, at the level of departments and activities, and not at the design of individual workstations (ININ 4040, Deck 02). Suppliers, the sterilizer, and the customer are treated as outside interfaces. They show up in the analysis as material coming in through receiving and going through shipping, but they are not part of what we are designing. *Figure 1* shows this boundary: everything between the receiving docks and the shipping docks is inside the scope.

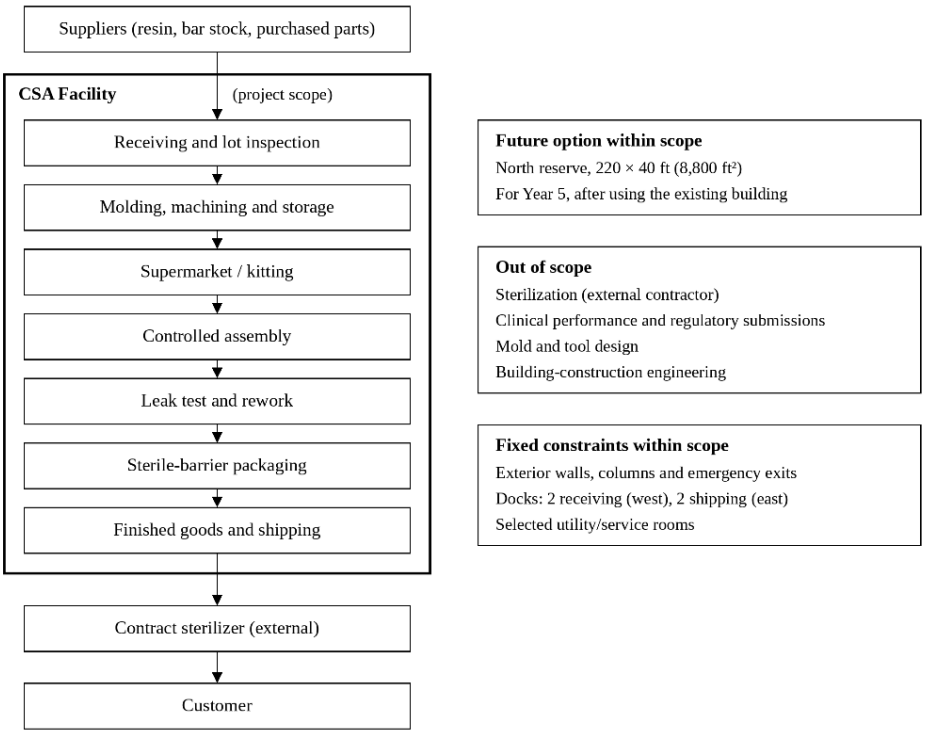


Figure 1. System boundary for the facility design problem.

Stakeholders

- CSA management and owners, who care about investment, capacity and growth.
- Production staff and operators, who are affected by ergonomics, walking distances, congestion, and safety.
- Quality and metrology, who are responsible for inspection, segregation, lot traceability and keeping the assembly area under control.
- Maintenance and tooling, who need access to the presses, lathes and utilities.
- Logistics (receiving, storage and shipping), as well as suppliers and contract sterilizers.

Requirements

ID	Requirement	Source
RQ-1	Produce 266,400 good devices/year in Year 1 (T5 132,000; T10 86,400; T12 48,000)	CSA Table 1, Section 3
RQ-2	Achieve it on the normal schedule: 250 days, 1 shift, 420 net min/day; routine overtime does not count as capacity	CSA Table 5, Section 2f
RQ-3	Show a credible physical and capacity strategy for 325,000 units/year in Year 5	CSA Section 2b, Section 3
RQ-4	Make the 9 MAKE parts in-house (6 molded, 3 machined shafts) and receive the 9 BUY parts	CSA Table 2
RQ-5	Keep forklifts out of the controlled assembly area	CSA Section 2a
RQ-6	Package only first-pass conforming devices or devices recovered through rework	CSA Section 2j
RQ-7	If a second shift is proposed, include its staffing and supporting services	CSA Section 2f

Table 2. Design requirements

Constraints

The design has to work within the existing 220 ft by 120 ft building. The exterior walls, the columns (20 ft east–west by about 17 ft north–south), the emergency exits, and some of the utility and service rooms cannot be moved. The same is true for the docks, which stay where they are: two for receiving on the west side and two for shipping on the east side.

There is a 220 ft by 40 ft expansion reserve on the north side, but it is limited, and the case asks that the current building be used first. On the equipment side, the T12 cannula mold has to run on the larger 140-ton press. Apart from that, most equipment and the non-structural interior walls can be moved if the change is justified (CSA, Section2a, Section 2g and Section 3).

Performance Criteria

When we start comparing alternatives later in the project, we plan to look at capacity against demand (utilization of each resource, in %), material-flow distance (ft/day), how many times the main flow crosses or backtracks, required space against available space (ft²), material-handling cost (USD/year), relocation and investment cost (USD), and how easily each option can grow to Year 5, which we will judge qualitatively.

Year 1 versus Year 5

Aspect	Year 1: requirement	Year 5: planning consideration
Demand	266,400/year (1,066/day)	325,000/year (1,300/day)
Mix T5 / T10 / T12	49.5% / 32.4% / 18.0%	44.6% / 32.3% / 23.1%
Required pace (takt)	23.6 s per unit	19.4 s per unit
Nature	Mandatory: the facility must meet it on one shift	Credible path: may rely on expansion, a second shift or phasing
Space	Within the existing building	May consider the north reserve

Table 3. Year 1 versus Year 5

The takt values come from 420 minutes per day and 250 days per year. It is also worth noting that the mix moves toward T12, which has the longest process times and the lowest first-pass test conformance (94.5%). Because of this, the workload grows faster than the number of units.

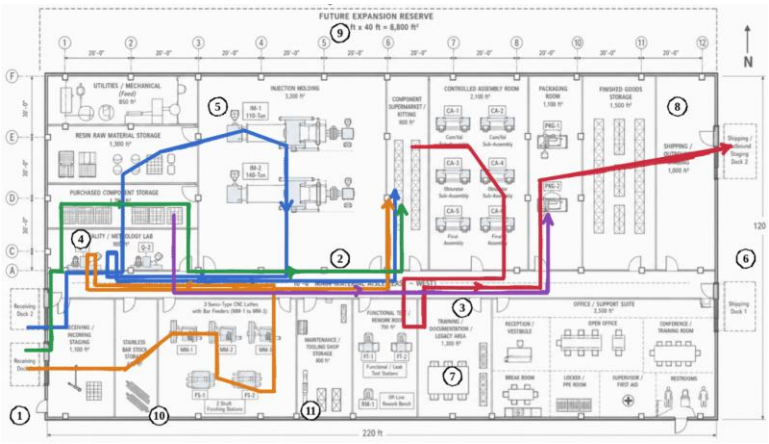
Definition Assumptions

To define the problem, we had to assume a few things: that the Table 1 demand and mix stay steady throughout each year, and that T12 goes through the same activities as T10. These assumptions are tracked in the register in Section X (E-02 and E-14).

Formal Problem Statement

CSA's Mayagüez facility currently makes 192,000 T5 and T10 trocars a year in a 26,400 ft² building on one shift. In Year 1 it has to make 266,400 good units and add the new T12, and by Year 5 that number rises to 325,000. If nothing changes, the facility may not be able to keep up with this volume and variety under normal operation. The facility therefore needs to handle Year-1 demand on one shift, inside the existing building and without touching its fixed elements, while also giving a credible path to Year 5 with efficient and controlled material flow.

III. Existing facility and production-system understanding



Material flows (approximate paths over case Fig. 3)

- A. Molded: resin > molding > inspection > supermarket
- B. Shafts: bar > CNC > finish > inspection > supermarket
- C. Purchased: receiving > inspection > storage > kitting
- D. Device: assembly > test > packaging > FG > dock
- E. Pouches and labels: west storage > packaging

1 Fixed receiving (W) and shipping (E): west-to-east flow
2 One 10-ft aisle for forklifts, pallet jacks and carts
3 Assembly (N) > test (S) > packaging (N): 2 crossings, backtrack
4 Quality lab in NW corner inspects 3 streams
5 Molding 3,200 ft², next to resin and utilities
6 Shipping dock 1 opens onto the aisle by the offices
7 Training/legacy area, 1,300 ft², in production zone
8 FG 1,500 + outbound staging 1,000 ft² by dock 2
9 8,800 ft² north reserve borders the north bay
10 Bar stock 650 ft² and CNC near receiving
11 Shaft cleaning has no location on the plan

Figure 2. Annotated Year-0 layout. The paths are approximate and follow the main aisle; the circled numbers match the observations in Section 3.6.

The current (Year 0) facility is a one-story building divided by a single 10-ft aisle that runs east to west. North of the aisle are the storage rooms, molding, kitting, the controlled assembly room, packaging and finished goods. South of it are receiving, machining, maintenance, functional testing and the offices (CSA, Figure 3). The three component streams (molded parts, machined shafts and purchased parts) all come together at the supermarket/kitting area, and from there the device goes through assembly, testing, packaging and shipping to the sterilizer (CSA, Figure 4). In general, material moves from west to east, but the device has to cross the aisle twice between assembly, testing and packaging.

Principal Activities and Their Location

Zone / department	Area (ft²)	Side	Year-0 resources	Role in the system
Receiving / incoming staging	1,100	South, west	Receiving docks 1 and 2	Entry of resin, bar stock, purchased parts and packaging
Stainless bar-stock storage	650	South, west	Bar racks	Shaft raw material
Swiss CNC machining and finishing	1,535	South	MM-1 to MM-3 (CNC), FS-1 and FS-2 (finishing)	Turning, cutoff and finishing of shafts
Maintenance / tooling shop	800	South	Benches and shelving	Support for presses and lathes
Functional test / rework	700	South, center	FT-1, FT-2, RM-1	Leak test and off-line rework
Training / documentation / legacy	1,300	South, center	Tables	Support; productive use not defined

Office / personnel services	2,500	South, east	Offices, lockers/PPE, break room, restrooms, first aid	Administrative and personnel support
Utilities / mechanical	850	North, west	Feed equipment	Selected fixed service room
Resin storage	1,300	North, west	Gaylords, drums	Molding raw material
Purchased-component storage	1,200	North, west	Shelving	Seals, stopcocks, springs, pouches, labels
Quality / metrology lab	900	North, west	Q-1, Q-2	Dimensional inspection of components
Injection molding	3,200	North	IM-1 (110 t), IM-2 (140 t), 2 dryers	Six MAKE plastic parts
Supermarket / kitting	1,350	North, center	Racks	Convergence point of the three streams
Controlled assembly	2,100	North, center	CA-1/CA-2 cannula, CA-3/CA-4 obturator, CA-5/CA-6 final	Parallel subassemblies and final assembly
Packaging	1,392	North, east	PKG-1, PKG-2	Sterile-barrier pouch and label
Finished goods	1,500	North, east	Racks	Pre-sterilization storage
Shipping / outbound staging	1,000	North, east	Shipping dock 2	Outbound to the sterilizer
Main aisle	2,200 (10 × 220)	Center	Forklifts, pallet jacks, carts	Circulation for all streams

Table 4. Principal activities and their location

If we add up all the labeled areas, the 17 rooms in the shell add up to about 23,545 ft². With the 10 ft aisle (about 2,200 ft²), the total is about 25,745 ft² of the 26,400 ft². That leaves only about 650 ft² for walls and door clearances. Machining is 1,535 ft². The production equipment on the plan matches Table 4 of the case, with 12 types of resources and 22 units in total.

Character of the Production System

Overall, this is a medium-to-high-volume system with low variety, since it is one product family in three sizes that share the same architecture. Molding and machining are grouped by process and run in batches, with mold or program changes between sizes. Assembly, testing and packaging are each in their own room, with two stations per operation. This describes how the plant works today (Year 0) and should not be read as a recommendation for what type of layout to use in the future.

Material Streams

Stream	Current route in the plant	Physical form (interpretation)	What the plan shows
A. Molded components	Receiving, resin storage, drying, molding, degate/trim, inspection, supermarket	Resin in gaylords or on pallets; parts in totes	Resin storage and molding are adjacent; inspection is in the lab at the northwest corner, opposite the direction of the supermarket
B. Machined shafts	Bar stock, bar feeder, Swiss CNC, finishing, cleaning, inspection, partitioned tray, supermarket	Bars on racks; shafts in partitioned trays	Bar stock and CNC are next to receiving; shafts cross the aisle to the lab and then travel to the supermarket. Rework repeats finishing and inspection, so it repeats that trip
C. Purchased components	Receiving, lot inspection, purchased storage, supermarket/kitting	Supplier cartons or pallets	Short route to storage; then about 70 ft along the aisle to kitting (measured on the figure)

D. Final device	Kitting, parallel subassemblies, final assembly, test, packaging, finished goods, shipping	Kits and WIP in totes/carts; finished goods in cases and on pallets	Leaves the controlled room (north), crosses to the test room (south, west of assembly) and crosses back to packaging (north, east)
E. Packaging materials	Receiving, lot inspection, storage, packaging staging, packaging	Pouches and labels in cartons	No packaging-material staging area appears on the plan, so pouches and labels have to travel almost the entire length of the building, from west to east.

Table 5. Material streams

Material-Handling Practices

1. Forklifts handle the docks and palletized bulk storage, but they are not allowed inside the controlled assembly area (CSA, Section 2a).
2. Pallet jacks are used for some of the internal pallet moves.
3. Carts and totes or trays carry work in process and replenish the lines.
4. All of this equipment uses the same 10-ft aisle, which is also the only connection between the north and south sides of the building. The case does not give trip frequencies, load sizes or assigned routes, so we added these to the register as information we still need (E-09).

Physical Constraints

1. The fixed elements are the exterior walls, the columns (on a grid of about 20 ft east–west by 30 ft north–south), the docks (two receiving docks on the west and two shipping docks on the east), the emergency exits and some of the utility and service rooms (CSA, Section 3).
2. Most of the equipment and the interior partitions that are not structural can be moved if the design justifies it.
3. We still need to confirm which service rooms are fixed (for now we assume Utilities/Mechanical is), whether the presses have foundations or utility connections that

make them hard to move, and what environmental conditions the controlled room requires (E-08, E-10 and E-11)

4. We also noticed some inconsistencies in the drawing. The grid goes from line 8 straight to line 10, its dimensions add up to 200 ft even though the building is 220 ft long, and the shell spaces rows A to G at about 17 ft (7 × 17 ≈ 120 ft), and the row label G is used twice. Since the figure seems to be schematic. We take dimensions from the annotated shell.

Observations on the Current Layout

- The numbers below match the circled numbers in *Figure 2.* These are things we observed in the current layout, not proposals for a new one.
- Since receiving is on the west side and shipping is on the east side, material generally flows from west to east.
- One 10-ft aisle carries all five material streams, along with forklift, pallet jack and cart traffic.
- The device goes from assembly (north side) to testing (south side, and further west), then back across to packaging (north side, further east). This means two aisle crossings and one backtrack.
- The quality lab is in the northwest corner and handles inspection for molded parts, shafts and most likely purchased lots as well (E-12).
- Molding is the largest department at 3,200 ft². It sits between resin storage and utilities on the west and the supermarket on the east.
- Shipping dock 1 opens onto the east end of the aisle, right next to the offices, instead of onto the outbound staging area.
- The training/documentation/legacy area (1,300 ft²) takes up space in the production zone, and its use is not clearly defined.
- Finished goods (1,500 ft²) and outbound staging (1,000 ft²) are next to each other and next to dock 2.
- The 8,800 ft² expansion reserve runs along the whole north wall, next to storage, molding, kitting, assembly, packaging and finished goods.
- Bar stock and machining are close to receiving, so incoming bars do not have to travel far.

- Shaft cleaning is part of the shaft route, but the plan does not show where it happens (E-13).

IV. Product Information

According to Muther's Systematic Layout Planning framework (ININ 4040, Deck 04), PQRST provides the essential information to gather before starting a meaningful facility analysis. The first category is Product (P), defining what the facility needs to produce, receive, and handle. For this project, the product information comes from the product family BOM (CSA, Table 2, p. 5) and the exploded view in Figure 1 (CSA, p. 1) of the case description. This information is crucial for understanding the project's components. *Figures 3-5* present this information in a detailed Product worksheet in Excel. The initial draft was prepared with the assistance of Claude (Anthropic, 2026) and was then reviewed and edited by the team. The worksheet distinguishes between what the case states and the team's interpretation and its facility implications. Overall, the worksheet shows that T12 adds a new size to the product family, not a new product architecture. It introduces three new part numbers and shares all its other components with T10 or with the whole family. The following paragraphs explain what this product structure means for the facility: part identification, material streams, and the information still to be verified.

P – PRODUCT-DERIVED FACILITY INFORMATION WORKSHEET (Preliminary)

Source: CSA Semester Project Case Description Fall2026 v0.2 (Sec. 1, 2a, 2c, 2g, 2k; Fig. 1; Tables 2–4)
Purpose: Derive facility-design relevant information from controlled product definition.
Note: ***Does not include process or quantity information. To be matured with R, Q, S and T.***

Status Legend (Muther)	
CURRENT	Directly supported by controlled product evidence.
ASSUMED	Engineering assumption made due to incomplete information.
VERIFY	Requires confirmation before being treated as established.

ID	Controlled Product Evidence (from BOM)	Facility-Design Interpretation	Preliminary Facility Implication / Requirem…	Status (Muther)	Verification Need / Note…	Source Reference
P-1	CSA Trocar Family: T5, T10, T12 (new)	T5 (Ø5 mm) and T10 (Ø10 mm) are currently in production; T12 (Ø12 mm) is to be introduced. The three trocars have the same architecture (cannula assembly + obturator assembly + packaging).	To ensure configuration control, the facility needs to support 3 variants simultaneously during all stages: production, assembly, handling, and storage.	CURRENT	It's important to confirm with R that T12 will not bring in any new operation types, just new part numbers.	Sec. 1, p. 1; Fig. 1, p. 1
P-2	CSA BOM — 18 part numbers (9 MAKE / 9 BUY)	There are 6 molded polymer (PL) parts and 3 machined stainless (MT) MAKE parts, alongside 5 purchased components (PU) and 4 packaging items (PK). There are two internal production streams: one for polymer and one for metal, along with purchased-component and packaging streams.	For MAKE items, the facility must have the capacity for internal production, while BUY items require receiving, verification of incoming items, controlled storage, and presentation.	CURRENT	None at this time.	Table 2, p. 5
P-3	Molded polymer components — MAKE items	The cannula bodies, seal housings, and obturator handle are all molded in-house from resin. The cannula body is made of transparent medical-grade polycarbonate, with its surfaces visible on the finished device.	This requires resin storage, molding capability, and WIP storage for molded components. Molded parts require careful handling to avoid scratches or contamination.	CURRENT	It's assumed that the transparent cannula has a certain surface sensitivity; make sure it meets cosmetic and handling requirements.	Table 2, p. 5; Sec. 1, p. 2; Fig. 1, p. 1
P-4	Cannula bodies — MAKE (size-specific; T12 new)	PL-100-05/10/12 is available in one size. PL-100-12 represents a new component, utilizing a larger mold allocated to the bigger press class, which is the 140-ton class, the more powerful of the two existing presses.	With T12, the press class gains additional load and a new mold, as well as implications for mold storage and changeover space.	CURRENT	Mold-to-machine compatibility for the other molded parts will be supplied in the project data package.	Table 2, p. 5; Sec. 2g, p. 8; Table 4, p. 7
P-5	Seal housings — MAKE (PL-200-U shared T10/T12)	PL-200-05 exclusively uses T5, while PL-200-U is compatible with both T10 and T12. T12 increases PL-200-U volume rather than adding a new part number.	There is no designated storage location for T12's seal housing; instead, the increased volume of PL-200-U must be stored and staged.	CURRENT	None at this time.	Table 2, p. 5

Figure 3. Product-Derived Facility Information Worksheet (P-1 to P-5)

ID	Controlled Product Evidence (from BOM)	Facility-Design Interpretation	Preliminary Facility Implication / Requirement	Status (Muther)	Verification Need / Note	Source Reference
P-6	Obturator handle PL-300 — MAKE (common)	This molded polymer MAKE part shares the highest commonality; its volume aligns with the overall family volume.	Single part number simplifies storage and kitting; its handling volume grows with total demand.	CURRENT	None at this time.	Table 2, p. 5; Fig. 1, p. 1
P-7	Obturator shafts/tips — MAKE (stainless, size-specific)	The MT-400-05/10/12 machine is made from stainless steel, available in one size each, and the MT-400-12 model is brand new. Keep the metal stream separate from the polymer parts; the tip is a functional, sharp component.	Bar-stock storage and machining capability are required. Shafts must be cleaned and protected (tip) before entering controlled assembly; contaminants from machining are kept away from controlled areas.	ASSUMED	Tip protection and contamination separation are engineering assumptions; confirm handling and cleanliness requirements.	Table 2, p. 5; Sec. 1, p. 2; Fig. 1, p. 1
P-8	Instrument and gas/check seals — BUY items	PU-500-05 (T5), PU-500-U (T10/T12) and PU-510 (all), silicone. Soft elastomer components are inspected on a lot basis.	Receiving, lot-based incoming inspection, controlled storage, and presentation to the cannula subassembly. Protection from deformation and contamination is crucial.	CURRENT	Confirm any special storage conditions for silicone seals (protection/shelf life).	Table 2, p. 5; Fig. 1, p. 1; Sec. 2k, p. 8
P-9	Stopcock and retaining spring — BUY (common)	PU-520 (polycarbonate/ABS) and PU-530 (stainless steel) are standard across all products, each with a single part number in the family.	For receiving, incoming verification, controlled storage, and kitting, no segregation by variant is necessary for these items.	CURRENT	None at this time.	Table 2, p. 5; Fig. 1, p. 1
P-	Pouch and labels — BUY packaging	The Sterile-barrier pouch PK-600 are common and labels PK-610-05/10/12 are size-specific. The pouch is the	Separate streams for packaging materials include receiving, inspection, clean/dry storage,		Confirm label control method (pre-printed vs. print-on-	

ID	Controlled Product Evidence (from BOM)	Facility-Design Interpretation	Preliminary Facility Implication / Requirement	Status (Muther)	Verification Need / Note	Source Reference
P-12	T12 introduction — 3 new part numbers	New items include PL-100-12, MT-400-12, and PK-610-12. T12 shares PL-200-U and PU-500-U with T10 and uses all common parts, while most purchased supplies and kitting content are reused.	New storage locations, a new mold, a new shaft program/tooling, and a new label are required. The existing BUY streams expand in volume rather than variety.	CURRENT	Volume impact to be matured with Q (Table 1).	Table 2, p. 5
P-13	Project boundary: external sterilization	Single-use devices packaged in a sterile-barrier pouch and sterilized by an external supplier after leaving the facility. The facility ships packaged, non-sterile devices; cleanliness still matters.	It has a controlled environment for assembly, test, and packaging (excluding forklifts from the controlled assembly area), along with pre-sterilization finished-goods storage.	VERIFY	Cleanliness/environmental classification of the controlled area is not stated in the case.	Sec. 1 (Project boundary), p. 2; Sec. 2a, p. 2
P-14	Product size: nominal diameters only	Fig. 1 provides nominal diameters (5, 10, 12 mm) for conceptual use only, without any lengths, weights, or package dimensions. Parts and devices appear small and lightweight.	Small-parts handling is consistent with current tote/tray/cart practice; unit-load and storage sizing cannot be determined yet.	VERIFY	Need part and package dimensions, weights, units per tote/case/pallet.	Fig. 1, p. 1
P-15	Seals: critical functional feature	Seals prevent loss of CO_2 used to inflate the abdomen. Sealing components (seal housings, instrument and gas/check seals) and their interfaces are critical to function.	It's crucial to protect sealing parts and surfaces during handling, storage, and presentation; this supports the need for functional/leak verification (see R).	ASSUMED	Critical surfaces/features are not specified; confirm with product specifications.	Sec. 1, p. 1; Fig. 1, p. 1
P-16	Material classes in BOM	Different materials like polymer resin (polycarbonate and other molded polymers), stainless bar stock, silicone, PC/ABS assemblies, pouches, and labels each have specific storage and handling requirements.	Separate storage conditions are likely necessary: resin requires moisture control/drying, bar stock is heavy and palletized, pouches and labels need clean, dry storage.	ASSUMED	Confirm storage conditions and incoming unit loads per material class.	Fig. 1, p. 1; Table 3, p. 6

Figure 5 (continued). Product-Derived Facility Information Worksheet (P-12 to P-16)

Of the 18-part numbers listed in the BOM, 11 are specific to sizes. These include the cannula bodies, the obturator shafts, the 5-mm instrument seal, and the product labels (Figure 4, P-11). Since these parts look the same across sizes, the facility needs to make sure they are correctly identified and kept separate during storage, kitting, and at every workstation to prevent mixing different versions. The label is the way to tell the different versions apart during packaging, so controlling the labels is very important. On the other hand, T12 brings in only three new part numbers and uses the rest from T10 or the whole family (Figure 5, P-12). In other words, the introduction of T12 mainly increases the number of parts, rather than the number of different parts. The product structure creates four material streams: two internal ones (molded polymer parts and machined stainless-steel shafts) and two purchased ones (components and packaging materials) (Figure 3, P-2); each of these streams has requirements. The shafts need to be cleaned and protected before they go into the controlled assembly area (Figure 4, P-7). The purchased items must go

through receiving, inspection based on lots, and controlled storage before they are used in assembly or packaging (Figure 4, P-8 to P-10). The T12 cannula mold requires a press, and the facility only has one of those (Figure 3, P-4). Lastly, even though sterilization happens outside the facility, the pouch is what keeps things sterile, so packaging still needs a controlled environment (Figure 5, P-13).

Although the case provides detailed product information, it falls short in some areas. The case gives nominal diameters with no lengths, weights or package dimensions (Figure 5, P-14), and the mold-to-machine compatibility for the remaining molded parts has not yet been released (Figure 3, P-4). The cleanliness classification of the controlled area (Figure 5, P-13), the critical sealing features (Figure 5, P-15), and the storage conditions for each material class (Figure 5, P-16) are also unconfirmed. All these product characteristics are marked as ASSUMED or VERIFY on the worksheet; therefore, they will be carried into the evidence and verification register in Section X.

V. Routing and process information

Production information alone does not define the production process. The product definition tells us specifically what must be produced, while routing tells us how production must occur. Therefore, facilities planning needs both views. Routing does not specify the layout, but it does provide evidence that the facilities designer must interpret to continue the detailed analysis.

CSA's production system includes five material streams (Case Table 3). Three of them, including molded components, machined shafts, and the final device, are MAKE or ASSEMBLE streams. Since CSA performs internal operations on them, each one is documented on its own manufacturing routing (See *Tables 6-8*). The remaining two are BUY streams, meaning that they do not require a manufacturing route because CSA does not process these items internally. However, they still require receiving, verification, storage, staging, and presentation at the point of use (Deck 04, slide 23). Following the BUY-item flow framework in Deck 04 (slide 23), with component data from Case Tables 2–3.

Seq. No.	Operation Description	Work Center/ Capability	Setup Time	Run Time (T5/T10/T12)	Inspection /Control	Notes/Remarks
10	Resin Drying / Staging	Resin dryer/feed system (2)	---		---	Resin from resin storage
20	Injection Molding	110-ton or 140-ton press (1 each)	---	22 – 36 s/shot (2-4 cavities)	---	T12 cannula mold assigned to 140-ton press
30	Degate / Trim	Degate/trim operation	---		---	Removes gates
40	Visual / Dimensional Inspection	Component dimensional inspection station	---		Visual/Dimensional Inspection	1.0-3.0% post-mold scrap; no rework.

Table 6. Manufacturing Routing of Molded Components

Seq. No.	Operation Description	Work Center/ Capability	Setup Time	Run Time (T5/T10/T12)	Inspection/ Control	Notes/Remarks
10	Swiss CNC Automatic Turning/Cut off	Swiss-type CNC automatic lathe (3)	---	50 / 65 / 80 s	---	Bar stock from bar-feeder staging
20	Precision Finish	Shaft finishing station (2)	---	25 / 30 / 35 s	---	Repeated for reworkable shafts
30	Cleaning	Cleaning (minor support activity)	---		---	Not a capacity-constrained resource

40	Dimensiona l Inspection	Component dimensional-inspection station	---		Dimensiona l Inspection	4–6% nonconforming; reworkable shafts return to Seq. 20

Table 7. Manufacturing Routing of Machined Shafts

Seq. No.	Operation Description	Work Center/ Capability	Setup Time	Run Time (T5/T10/ T12)	Inspection/Control	Notes/Remarks
10	Kitting	Supermarket/kitting	—	—	—	MAKE and BUY components presented
20	Cannula Subassembly	Cannula subassembly bench (2)	—	0.65 / 0.80 / 0.95 min	—	Parallel with Seq. 30
30	Obturator Subassembly	Obturator subassembly bench (2)	—	0.55 / 0.65 / 0.75 min	—	Parallel with Seq. 20
40	Final Assembly	Final assembly bench (2)	—	0.45 / 0.55 / 0.65 min	—	Cannula and obturator join
50	Functional / Leak Test	Functional/leak-test station (2)	—	0.45 / 0.55 / 0.65 min	Functional / Leak Test	94.5–97.0% first-pass conforming
55	Off-Line Rework (alternate)	Off-line rework bench (1)	—	—	Rework verification	Recovered units rejoin at Seq. 60; failed units scrapped
60	Packaging	Packaging station (2)	—	0.55 / 0.60 / 0.65 min	—	Pouch and label added
70	Pre-Sterilization FG Storage / Shipping	FG storage / outbound (east docks)	—	—	—	Ships to contract sterilizer

Table 8. Manufacturing Routing of Final Device

The routings indicate various process interfaces that the facility must support. The molded, machined, and purchased streams meet at kitting, where the facility must organize and control components for three product sizes (Deck 04, slide 22). The cannula and obturator subassemblies operate concurrently and need to be aligned by product type prior to the final assembly (Case Table 3). Two routing processes also include rework loops that generate reverse flow: nonconforming shafts can undergo precision finishing and inspection multiple times (Case section 2h), and devices that fail functional or leak tests are reworked off-line before reintegrating into the flow at packaging (Case section 2j). Moreover, packaging materials link to the final-device process at the packaging stage, and the three specific labels need oversight to avoid confusion. Ultimately, assembly occurs in a regulated space where forklifts are prohibited, leading to a shift in material handling from pallets to carts and totes at its perimeter (Case section 2a). While T12 does not introduce new operations, it increases product variety and workload on current routes, especially on the 140-ton press designated for the T12 cannula mold (Case section 2g).

The BUY streams provide a full understanding of how materials are integrated into the production system. *Table 9* lists these items, the products they pertain to, and their points of use (Case Tables 2–3; Deck 04, slide 23). Purchased components are shown during kitting, whereas packaging materials skip kitting and are prepared directly for packaging (Case Table 3). In both processes, rejected supplier lots are isolated and sent back to the supplier (Case section 2k), indicating a hold activity within the facility.

Controlled BUY Component	Part No.	Applies To	Internal Routing	Point of Use
Instrument seals (5-mm, universal)	PU-500-05, PU-500-U	T5; T10/T12	None	Kitting (Seq. 10)
Gas/check seal	PU-510	All	None	Kitting (Seq. 10)
Insufflation stopcock assembly	PU-520	All	None	Kitting (Seq. 10)

Controlled BUY Component	Part No.	Applies To	Internal Routing	Point of Use
Retaining spring	PU-530	All	None	Kitting (Seq. 10)
Sterile-barrier pouch	PK-600	All	None	Packaging (Seq. 60)
Product labels	PK-610-05/10/12	T5; T10; T12	None	Packaging (Seq. 60)

Table 9. Controlled BUY Components and Facility Flow

Together, the routings and BUY flows provide the evidence for the facility activities the system must support. Following Deck 04 (slides 24–25), these were consolidated into the candidate activities shown in *Table 10*. Not every routing step became a separate activity; for example, storage points were treated as logistics activities, and the three scrap exits were grouped into one. These are not departments or layout blocks; they serve as input to the activity model in Section VIII.

Inbound & Storage	MAKE Processing	Staging & Assembly Supply	Assembly & Outbound
1 Receiving	6 Resin Drying	14 Molded-Component Supermarket	19 Cannula Subassembly
2 Incoming Lot Inspection	7 Injection Molding	15 Shaft Tray Supermarket	20 Obturator Subassembly
3 Quarantine / Hold	8 Degate / Trim	16 Bar-Feeder Staging	21 Final Assembly
4 Raw-Material Storage	9 Swiss CNC Turning	17 Kitting	22 Functional / Leak Test

Inbound & Storage	MAKE Processing	Staging & Assembly Supply	Assembly & Outbound
5 Purchased-Component & Packaging Storage	10 Precision Finishing	18 Packaging-Material Staging	23 Off-Line Rework
	11 Shaft Cleaning		24 Packaging
	12 Component Dimensional Inspection		25 Pre-Sterilization FG Storage
	13 Scrap Collection		26 Shipping / Outbound Staging

Table 10. Candidate Facility Activities

Routing items that still require verification, such as setup times, mold-to-press compatibility, shared inspection stations, and molding scrap disposition, are carried forward to the evidence register in Section X.

VI. Quantity and timing information

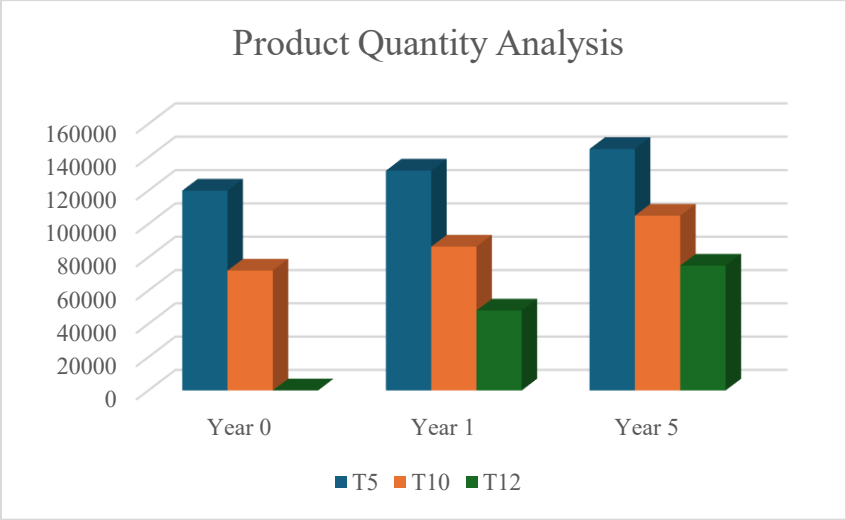


Figure 6. Annual Demand by Product (Years, 0, 1 and 5)

Caribe Surgical Access is expected to experience an increase in production demand during the five-year planning period. According to Table 1 of the case study, annual demand will increase from 192,000 units in Year 0 to 266,400 units in Year 1, representing a 38.75% increase. By Year 5, demand is expected to reach 325,000 units, which represents an overall increase of 69.27%. Based on 250 working days per year, the average daily demand increases from 768 units in Year 0 to approximately 1,066 units in Year 1 and 1,300 units in Year 5. This increase is significant because the facility must be able to support higher production volumes while maintaining product quality and efficient material flow.

The Product Quantity (P–Q) analysis presented in *Figure 6* shows the changes in demand for the three product models. T5 remains the product with the highest demand throughout the planning period, increasing from 120,000 units in Year 0 to 145,000 units in Year 5. However, its percentage of total demand decreases from 62.50% to 44.62%. T10 also increases in annual demand, while T12 is introduced in Year 1 with 48,000 units and reaches 75,000 units by Year 5. As a result, T12 represents 23.08% of total demand in Year 5. Since the three products share several manufacturing and assembly activities, these changes in product mix may increase the workload of shared equipment and require better coordination between production areas.

Operating Time and Equipment Availability	
Production Days per Year	250 days/year
Shifts	1 shift/day
Shift duration	8 hours
Net Operating Time	420min/day
Annual net operating time	1750 hours/year
Injectiong Molding Availability	90%
CNC availability	92%
Finishing availability	95%

Table 11. Operating Time and Equipment Availability

CSA operates 250 days per year with one 8-hour shift per day. After considering breaks, startup activities, and housekeeping, the facility has 420 minutes of net scheduled production time per

day, equivalent to 1,750 hours per year. Equipment availability is estimated at 90% for injection molding, 92% for CNC machining, and 95% for finishing equipment. These operating conditions must be considered when evaluating the production system, since equipment availability affects the time available for manufacturing. The facility must meet Year 1 demand under the normal one-shift schedule, while Year 5 requires further evaluation of future operating needs. Additional analysis will be necessary to identify possible workload limitations and determine how the facility can support increasing demand.

VII. Supporting services information

Service ID	Supporting Service	Activities Supported	Case Study Evidence	Facility Implications	Verification Needs
S-01	Quality / Metrology	Incoming inspection, component inspection, and production quality control	Table 3, Figure 3	Requires access between inspection and production areas.	Confirm inspection equipment and future requirements.
S-02	Maintenance / Tooling	Injection molding, CNC machining, and finishing equipment	Figure 3	Equipment must remain accessible for maintenance.	Verify maintenance access and tooling needs.
S-03	Utilities / Mechanical	anufacturing equipment and facility operatio	Figure 3	Equipment requires appropriate utility connections.	Confirm utility capacity and connection requirements.
S-04	Material Handling	Receiving, storage, production, and shipping	Table 3, Section 2	Materials must move safely between production areas.	Verify internal handling and movement requirements.
S-05	Training / Documentation	Production personnel and manufacturing operations	Figure 3	Requires access to training and documentation resources.	Confirm future training and personnel needs.
S-06	Off-line Rework	Functional testing and recovery of eligible rejected units	Tables 4 and 9	Requires a separate area and controlled movement of rework units.	Verify rework flow and handling requirements.

Table 12. Supporting Services and Facility Implications

CSA requires several supporting services to maintain its manufacturing operations. Based on the case study, these services include quality and metrology, maintenance and tooling, utilities, material handling, training and documentation, and off-line rework. Each service supports different production activities. Quality control helps ensure that components meet the required specifications, while maintenance supports the operation of manufacturing equipment. Utilities provide the necessary services for production, and material handling supports the movement of materials between receiving, storage, manufacturing, and shipping. Training and documentation support employees and production procedures, while off-line rework allows eligible devices that fail testing to be evaluated and corrected.

These supporting services are important when considering the facility layout because they must be connected to the production activities they support. As CSA introduces T12 and increases production demand, the workload of some supporting services may also increase. For example,

higher production volumes may require additional inspection, material movement, and equipment maintenance. However, the current information is not enough to determine whether additional resources or space will be needed. Further evaluation is required to verify the capacity, accessibility, and future requirements of these services before making final layout decisions.

VIII. Preliminary facility activity model

Activity ID	Activity Name	Classification	Evidence	PQRST Basis	Rationale	Conclusion Maturity	Verification Needs
A-01	Receiving	Primary	Table 3	P, R	Receives raw materials and purchased components.	Firm	Verify receiving and material handling requirements.
A-02	Resin Storage	Primary	Table 3	P, R	Stores resin before production.	Firm	Verify storage conditions and material quantities.
A-03	Resin Drying and Staging	Primary	Table 3; Table 4	P, R, T	Prepares resin for injection molding.	Firm	requirements.
A-04	Injection Molding	Primary	Tables 3, 4, 6	P, Q, R, T	Produces molded plastic components.	Firm	Verify equipment workload and T12 requirements.
A-05	Degating and Trimming	Primary	Table 3	P, R	Removes excess material from molded components.	Firm	Verify work area and handling requirements.
A-06	Inspection	Primary	Table 3	P, R, S	Checks molded components before storage.	Firm	Verify inspection requirements and product flow.
A-07	Bar-Stock Storage	Primary	Table 3	P, R	Stores metal bars used for shaft production.	Firm	Verify storage and material handling requirements.
A-08	Swiss-CNC Machining	Primary	Tables 3, 4, 7	P, Q, R, T	Produces metal shafts and tips.	Firm	Verify machine workload and T12 processing needs.
A-09	Precision Shaft Finishing	Primary	Tables 3, 4, 7	P, Q, R, T	Completes finishing operations on machined shafts.	Firm	Verify equipment workload and material flow.
A-10	Shaft Cleaning	Primary	Table 3	P, R	Removes residue from machined shafts.	Firm	Verify cleaning requirements.
A-11	Shaft Dimensional Inspection	Primary	Table 3	P, R, S	Checks shaft dimensions before assembly.	Firm	Verify inspection equipment and requirements.
A-12	Storage	Primary	Table 3	P, R	Stores purchased components before assembly.	Firm	Verify storage and inventory requirements.
A-13	Supermarket and Kitting	Primary	Table 3	P, Q, R	Organizes components needed for assembly.	Firm	Verify kitting and material flow requirements.
A-14	Cannula Subassembly	Primary	Tables 3, 4, 8	P, Q, R, T	Assembles cannula components.	Firm	Verify workload and product handling requirements.
A-15	Obturator Subassembly	Primary	Tables 3, 4, 8	P, Q, R, T	Assembles obturator components.	Firm	Verify workload and product handling requirements.
A-16	Final Assembly	Primary	Tables 3, 4, 8	P, Q, R, T	Combines components into finished devices.	Firm	Verify assembly workload and T12 requirements.
A-17	Functional and Leak Testing	Primary	Tables 3, 4, 9	P, Q, R, T	Checks device performance before packaging.	Firm	Verify testing workload and rework flow.
A-18	Packaging	Primary	Tables 3, 4, 8	P, Q, R, T	Packages devices that pass testing.	Firm	Verify packaging workload and product separation.
A-19	Finished Goods Storage	Primary	Table 3	P, Q, R	Stores packaged devices before shipping.	Firm	Verify storage and inventory requirements.
A-20	Shipping	Primary	Table 3	R, Q	Sends packaged devices to external sterilization.	Firm	Verify shipping and material handling requirements.
A-21	Quality and Metrology	Supporting	Table 3; Figure 3	S, R	Supports inspection and product quality activities.	Preliminary	Verify inspection support requirements.
A-22	Maintenance and Tooling	Supporting	Figure 3	S, T	Supports equipment operation and maintenance.	Preliminary	Verify maintenance and tooling needs.
A-23	Utilities and Mechanical	Supporting	Figure 3	S	Provides utility support for manufacturing equipment.	Preliminary	Verify utility capacity and connections.
A-24	Internal Material Handling	Supporting	Table 3; Section 2	S, R	Supports material movement between production areas.	Preliminary	Verify handling methods and movement requirements.
A-25	Training and Documentation	Supporting	Figure 3	S	Supports employee training and production documentation.	Preliminary	Verify future training requirements.
A-26	Off-line Rework	Supporting	Tables 4, 9	S, R, T	Handles eligible devices that fail functional testing.	Firm	Verify rework flow and handling requirements.
A-27	Incoming Inspection	Primary	Table 3	P, R, S	Checks purchased components before storage and produ	Firm	Verify inspection and quarantine requirements.
A-28	Packaging Material Staging	Primary	Table 3	P, Q, R	Prepares packaging materials before packaging operation	Firm	Verify staging and material handling requirements.

Table 13. Preliminary Facility Activity Model

The preliminary facility activity model was developed using the PQRST information provided in the case study. This information includes the products manufactured by Caribe Surgical Access (CSA), production quantities, manufacturing routes, supporting services, and operating times. Based on this information, a total of 28 activities were identified and organized into primary and supporting activities. Primary activities include receiving, material storage, injection molding, CNC machining, component inspection, assembly, functional testing, packaging, and shipping. Supporting activities include quality and metrology, maintenance, utilities, material handling, training, and rework. Each activity was assigned an identification number and documented with its corresponding evidence, PQRST basis, and purpose within the production system. This organization helps explain how the different activities contribute to the manufacturing process and how they are connected.

The activity model also identifies the level of certainty of each activity and the information that still needs to be verified. Activities directly identified in the case study were classified as firm, while some supporting-service requirements remain preliminary because additional information is needed. Important verification needs include equipment workload, material flow, inspection requirements, storage conditions, and the resources needed to introduce T12. Since production demand is expected to increase in Years 1 and 5, these factors should be evaluated before making decisions about the facility layout. Overall, the preliminary activity model provides a basis for understanding the current production system, identifying possible areas of concern, and supporting the next stages of the facility design process.

IX. Preliminary activity relationships

After defining the facility activities in the preliminary activity model, the next step is to determine which activities need to be close to each other and which should be separated, and specifically why. Each pair of activities was rated using Muther's closeness codes: A (absolutely necessary), E (especially important), I (important), O (ordinary), U (unimportant), and X (undesirable). A, E, and I mean the activities should be close, with A being the strongest; O and U mean their placement does not matter much; and X means they should be kept apart. The maturity of each relationship follows the same scale used in the verification register (Section X): Firm, Preliminary or Open. The relationships are documented in three tables: significant flow relationships (*Table 14*), significant non-flow relationships (*Table 15*), and unresolved relationships (*Table 16*). Only pairs rated A, E, I, or X are listed, since these are the ones that will shape the layout; pairs rated O or U were evaluated but do not require a specific placement.

Pair	Rating	Evidence	Rationale	Maturity
1 Receiving – 4 Raw-Material Storage	A	Resin and bar stock arrive by forklift at the west docks (CSA section 2a)	Heavy palletized loads; short forklift moves	Firm
1 Receiving – 2 Incoming Lot Inspection	E	Purchased lots are inspected on arrival (CSA, Table 3)	Every purchased lot	Firm

2 Incoming Lot Inspection – 5 Purchased & Packaging Storage	E	Accepted lots go to storage (CSA, Table 3)	Every accepted lot	Firm
2 Incoming Lot Inspection – 3 Quarantine/Hold	E	Rejected lots are held and returned (CSA Section 2k)	Rejected lots must be isolated right away	Firm
4 Raw-Material Storage – 6 Resin Drying	A	Resin is dried before molding (CSA, Table 3)	Continuous resin supply	Firm
6 Resin Drying – 7 Injection Molding	A	Dryers feed the presses (CSA, Table 3)	Dried resin must reach the press directly	Firm
7 Injection Molding – 8 Degate/Trim	A	Every shot is degated (CSA, Table 3)	Immediate next step for every part	Firm
8 Degate/Trim – 12 Component Inspection	E	All molded parts are inspected (CSA, Table 3)	Every molded part	Firm
4 Raw-Material Storage – 16 Bar-Feeder Staging	A	Bars move from racks to the feeders (CSA, Table 3)	Long bar stock is hard to move far	Firm
16 Bar-Feeder Staging – 9 Swiss CNC	A	Feeders supply the lathes (CSA, Table 3)	Direct supply	Firm
9 Swiss CNC – 10 Precision Finishing	A	Every shaft is finished (CSA, Table 3)	Immediate next step	Firm
10 Precision Finishing – 11 Shaft Cleaning	E	Shafts are cleaned after finishing (CSA, Table 3)	Every shaft	Firm
11 Shaft Cleaning – 12 Component Inspection	E	Shafts are inspected after cleaning (CSA, Table 3)	Every shaft	Firm
12 Component Inspection – 10 Precision Finishing	E	4–6% of shafts return for rework (CSA Section 2h)	Rework loop repeats the trip	Firm
12 Component Inspection – 14 Molded Supermarket	E	Accepted molded parts go to the supermarket (CSA, Table 3)	Every accepted part	Firm
12 Component Inspection – 15 Shaft Tray Supermarket	E	Accepted shafts go to the supermarket (CSA, Table 3)	Every accepted shaft	Firm
14 Molded Supermarket – 17 Kitting	A	Molded parts are presented at kitting (CSA, Table 3)	Feeds every kit	Firm
15 Shaft Tray Supermarket – 17 Kitting	A	Shafts are presented at kitting (CSA, Table 3)	Feeds every kit	Firm
5 Purchased Storage – 17 Kitting	E	Seals, stopcocks and springs used at kitting (Table 2)	Frequent replenishment; today about 70 ft away	Firm

5 Purchased Storage – 18 Packaging-Material Staging	E	Pouches and labels skip kitting (CSA, Table 3)	Today they travel almost the full building	Firm
18 Packaging-Material Staging – 24 Packaging	A	Pouches and labels used at every unit (CSA, Table 3)	Point-of-use supply	Firm
17 Kitting – 19 Cannula Subassembly	A	Every kit enters assembly (CSA, Table 3)	Transfer from pallets to carts at the controlled room	Firm
17 Kitting – 20 Obturator Subassembly	A	Every kit enters assembly (CSA, Table 3)	Same as above	Firm
19 Cannula Sub. – 21 Final Assembly	A	Cannula joins obturator (CSA, Table 3)	Every unit	Firm
20 Obturator Sub. – 21 Final Assembly	A	Obturator joins cannula (CSA, Table 3)	Every unit	Firm
21 Final Assembly – 22 Functional/Leak Test	A	100% of devices are tested (CSA, Table 3)	Highest-volume device move; today it crosses the aisle (Section 3.6)	Firm
22 Functional/Leak Test – 24 Packaging	A	94.5–97% pass first test (CSA, section 2j)	Nearly all units; today it crosses the aisle again and backtracks	Firm
22 Functional/Leak Test – 23 Off-Line Rework	E	3–5.5% fail first test (CSA, section 2j)	Return loop for every failed unit	Firm
23 Off-Line Rework – 24 Packaging	I	Recovered units rejoin at packaging (CSA section 2j)	Low volume	Firm
8, 12, 23 – 13 Scrap Collection	I	Molding scrap (1–3%), shaft and device scrap	Low volume; needs its own exit route	Preliminary
24 Packaging – 25 FG Storage	E	All packaged devices go to FG (CSA, Table 3)	Every unit	Firm
25 FG Storage – 26 Shipping	A	FG ships from the east docks (CSA, Table 3)	Palletized; must be next to the dock	Firm

Table 14. Significant flow relationships

Table 15 shows significant non-flow relationships. Unlike flow relationships, these do not depend on material movement. Instead, they come from the need to protect the controlled area, keep good products apart from scrap and rejected material, share supervision or equipment, and give support services access to the activities they serve.

Pair	Rating	Evidence	Rationale	Maturity
1 Receiving, 26 Shipping – 19, 20, 21 Assembly	X	Forklifts are not allowed in the controlled area (CSA, Section 2a)	Keep forklift traffic away from the assembly room	Firm
9 Swiss CNC, 10 Precision Finishing – 19, 20, 21 Assembly	X	Shafts must be cleaned before entering assembly (Figure 4, P-7; E-34)	Oils, chips and coolant could contaminate the room	Preliminary
7 Injection Molding – 19, 20, 21 Assembly	X	Controlled-area protection (CSA, Section 2a)	Heat, particles and noise from the presses	Preliminary
13 Scrap Collection – 17 Kitting, 21 Final Assembly, 24 Packaging	X	Only conforming or reworked devices may be packaged (CSA, Section 2j)	Scrap must never mix with good product	Firm
3 Quarantine/Hold – 17 Kitting	X	Rejected lots are held and returned to the supplier (CSA, Section 2k)	Rejected material must not reach kitting	Firm
19 Cannula Sub. – 20 Obturator Sub.	E	Subassemblies run in parallel and must match by size (CSA, Table 3)	Shared supervision and size alignment before final assembly	Firm
21 Final Assembly – 24 Packaging	E	The pouch keeps the device sterile (Figure 5, P-13)	Both need controlled conditions	Preliminary
18 Packaging-Material Staging – 24 Packaging	E	Three size-specific labels (Figure 4, P-11)	Label staging must stay at packaging to avoid mix-ups	Firm
7 Injection Molding – Maintenance/Tooling	E	Presses need mold and tooling support (Section VII)	Mold changes between sizes	Firm
6 Resin Drying, 7 Injection Molding – Utilities	E	Dryers and presses need power and cooling (Section VII; E-45)	Utility connections	Firm
9 Swiss CNC – Maintenance/Tooling	I	Lathes need maintenance (Section VII)	Tool changes and repairs	Firm
19, 20, 21 Assembly – Personnel Services	I	Controlled-room access (CSA, Section 2a)	Operators need gowning/PPE before entering	Preliminary

Table 15. Significant non-flow relationships

Table 16 lists the relationships that are still unresolved. For each one, it explains why the rating cannot be confirmed yet, what information would decide it, and where it is tracked in the verification register. These relationships are considered Open and will be revisited as new information is released.

Pair	Why it is unresolved	What would decide it	Register ID
All flow pairs	Ratings are based on the share of product on each route, not on real trip counts	Trip frequencies, unit loads and lot sizes	E-09
Controlled room vs. 7 Molding, 9 Swiss CNC, Maintenance, 22 Test, 24 Packaging, Personnel	Separation ratings are team assumptions, and it is unclear whether test and packaging must be inside the controlled area or whether gowning is required	Controlled-room requirements	E-11, E-34, E-42
11 Shaft Cleaning – 19, 20, 21 Assembly	Cleaning location is not shown on the plan	Confirm location and services	E-13
5 Purchased Storage – 18 Packaging-Material Staging	Staging area is not shown on the plan	Confirm whether it exists and where	E-21
2 Incoming Lot Inspection – 12 Component Inspection	Lot inspection is assumed to happen in the quality lab	Confirm where lot inspection is done	E-12
6 Resin Drying, 7 Injection Molding – Utilities	Dryer location and press utility needs are not confirmed	Dryer location, press utilities and whether presses can move	E-22, E-10, E-45

Table 16. Unresolved relationships

The ratings point to three main conflicts with the current layout. First, assembly, test, and packaging have A flow relationships, but today the device crosses the main aisle twice and backtracks between them (Section 3.6). Second, packaging materials have an E relationship with packaging, yet they are stored at the west end of the building. Third, the quality lab has E relationships with molding, machining, and kitting, but it sits in the northwest corner, so reworked shafts repeat the longest trip. The X relationships confirm that the controlled room must stay away

from forklift traffic, machining, and scrap, which the current layout partly achieves. Once the unresolved items are confirmed, the from-to chart in the next stage will quantify flow intensity.

X. Evidence assumptions and verification register

To keep track of the information used throughout this report, the team built an evidence, assumptions and verification register in Excel (*Figures 7 to 12*). Each entry records the item, its source, its evidence status, our current interpretation, what still needs to be verified, and how the analysis would change if that item turned out to be different. Following the course approach (ININ 4040, Deck 04), each item is marked as CURRENT when it comes directly from the case data, ASSUMED when it is a reasonable assumption we have not yet confirmed, or VERIFY when it must be confirmed before it can be relied on. We also rated the maturity of the conclusion drawn from each item (Firm, Preliminary or Open), since an item backed by case data can still lead to a conclusion that is only preliminary. The register currently holds 45 entries covering all ten sections: 19 CURRENT, 9 ASSUMED and 17 VERIFY. Most of the VERIFY items relate to physical data from the floor plan, unit loads, mold changeovers, and staffing. Because these are the items most likely to change the activity relationships and the space the facility needs, they should be confirmed first. Each entry keeps its ID (E-01 to E-45) as it is updated, so that changes can be traced back to their source as new information is released.

ID	Area	Item	Source / Evidence	Status	Current Interpretation	Verification Need	Impact if Changed	Maturity
E-01	Problem definition, existing facility, assessment	Building envelope: 220 × 120 ft (26,400 ft²); north reserve 220 × 40 ft (8,800 ft²)	CSA §2a, §3; Fig. 2	CURRENT	Year 1 is solved within the building; the reserve is an option for Year 5	None	Changes the entire space analysis	Firm
E-02	Problem definition, existing facility, assessment	Demand is stable within each year, with no seasonality	Team assumption; CSA Table 1 gives annual totals only	ASSUMED	Daily rate = annual demand / 250 days	None	Peaks would require more capacity or finished-goods inventory	Preliminary
E-03	Problem definition, existing facility, assessment	Good-unit demand: Y0 192,000; Y1 266,400; Y5 325,000, by product	CSA Table 1	CURRENT	Capacity basis; the mix shifts toward T12 (18% to 23%)	Check for errata in each new release	Changes the load on every resource	Firm
E-04	Problem definition, existing facility, assessment	Operating basis: 250 days, 1 eight-hour shift, 420 net min/day, 1,750 h/year; availability 90% molding, 92% CNC, 95% finishing	CSA Table 5	CURRENT	Available time per resource for Years 0, 1 and 5	Confirm whether 95% applies to manual stations	Changes effective capacity	Firm
E-05	Problem definition, existing facility, assessment	Each manual station has one operator and 1,750 h/year, with no further allowance	Team assumption for the Appendix A check	ASSUMED	Manual loads are compared against 1,750 h per station	Confirm with the data release (staffing, allowances)	With allowances, utilization rises	Preliminary
E-06	Problem definition, existing facility, assessment	Mold–press compatibility: only the T12 cannula is known to require the 140-ton press	CSA §2g	VERIFY	The other 5 parts are assumed moldable on either press	Confirm mold–press compatibility in the next data release	Could create a bottleneck at IM-2	Open

Figure 7. Evidence Assumptions and Verification Register (E-01-E-06))

E-07	Problem definition, existing facility, assessment	Plan dimensions: grid line 9 is missing, dimensions add to 200 ft, machining has no area	CSA Figs. 2 and 3; team DWG	VERIFY	The figure is schematic; ~3,100 ft² unlabeled	Measure in the DWG	Changes available areas and distances	Open
E-08	Problem definition, existing facility, assessment	Which service rooms are fixed	CSA §2a, §3 ("selected utility/service areas")	VERIFY	Utilities/Mechanical (850 ft²) is assumed fixed	Request the list of fixed rooms	Limits what can be relocated	Open
E-09	Problem definition, existing facility, assessment	Trip frequencies, unit loads and lot sizes	Not given in the case	VERIFY	Only the type of handling equipment is known	Request trip and unit-load data, or define assumptions for the from-to chart	Needed for flow relationships and distances	Open
E-10	Problem definition, existing facility, assessment	Whether the presses can be moved (foundations, cooling water, air, power)	CSA §2a (equipment mostly relocatable)	VERIFY	Treated as relocatable at a cost	Ask for the presses' utility requirements	A fixed press anchors molding	Open
E-11	Problem definition, existing facility, assessment	Environmental requirements of the controlled room, and whether packaging is also controlled	CSA §2a (no forklifts in assembly)	VERIFY	Assembly is controlled; packaging is assumed the same because the pouch is the sterile barrier	Request the room class or requirements	Changes separations, access and relationships	Open
E-12	Problem definition, existing facility, assessment	Location of lot inspection for purchased and packaging materials	CSA §2d, §2k; not shown in Fig. 3	ASSUMED	Assumed to be in the quality lab (Q-1, Q-2)	Confirm location and equipment	Changes the receiving–lab relationship	Preliminary
E-13	Problem definition, existing facility, assessment	Location of shaft cleaning	CSA Table 3, note in §2e	VERIFY	Minor activity with no modeled capacity, but no place on the plan	Confirm location and services (chemicals, water)	May need an area with services	Open
E-14	Problem definition, existing facility, assessment	T12 follows the same routes and activities as T10	routes); Table 2 (parts shared by T10/T12)	ASSUMED	T12 adds load, not new activities	Confirm with the next data release	A different route would add activities	Preliminary

Figure 8. Evidence Assumptions and Verification Register (E-07-E-14))

E-15	Problem definition, existing facility, assessment	Rework: shafts repeat finishing and inspection; devices go to off-line rework and rejoin after test	CSA §2h, §2j	CURRENT	There are return flows for shafts and devices	None	Affects finishing–lab and test–rework–packaging relationships	Firm
E-16	Problem definition, existing facility, assessment	Devices that fail rework are scrapped whole; only good product uses a pouch and label	CSA §2j	CURRENT	Components must cover test losses	None	Changes component quantities	Firm
E-17	Problem definition, existing facility, assessment	Rejected purchased lots are quarantined and returned	CSA §2k	CURRENT	A quarantine area is needed; it is not shown on the plan	Confirm where quarantine is located	Adds a controlled space near receiving	Preliminary
E-18	Problem definition, existing facility, assessment	Overtime is not capacity; a second shift is an alternative that needs staffing and services	CSA §2f	CURRENT	Year 1 is evaluated on one shift	None	A second shift changes personnel services	Firm
E-19	Problem definition, existing facility, assessment	Use of the training/documentation/legacy area (1,300 ft²)	CSA Fig. 3	VERIFY	Non-productive use; potential space to recover	Confirm whether it must be kept	Changes available space	Open
E-20	Problem definition, existing facility, assessment	Frequency of shipments to the sterilizer	CSA §1 and Fig. 4 (no frequency)	VERIFY	Finished goods accumulate between shipments	Request shipment frequency and size	Changes finished-goods storage	Open
E-21	Problem definition, existing facility, assessment	Packaging-material staging area	CSA Table 3; not shown in Fig. 3	VERIFY	Pouches and labels travel from west storage to east packaging	Confirm whether it exists and where	Affects the storage–packaging relationship	Open
E-22	Problem definition, existing facility, assessment	Location of the 2 resin dryer/feed systems	CSA Table 4; Fig. 3 ("Feed")	ASSUMED	Assumed next to each press, fed from Utilities	Confirm in the DWG	Affects the resin–molding–utilities relationship	Preliminary
E-23	Product, routing, relationships	Product family: T5, T10 and T12 share one architecture (cannula assembly, obturator assembly, packaging)	CSA §1, Fig. 1, Table 2	CURRENT	T12 is a new size, not a new product; the same activity set applies	None	A new architecture would add activities	Firm

Figure 9. Evidence Assumptions and Verification Register (E-15-E-23)

E-24	Product, routing, relationships	MAKE (6 molded, 3 machined shafts) and 9 BUY; T12 adds 3 unique parts (PL-100-12, MT-400-12, PK-610-12)	CSA Table 2	CURRENT	Part numbers in use grow from 15 to 18; at kitting from 12 to 14 (the T12 label goes to packaging)	None	Changes storage locations and kitting variety	Firm
E-25	Product, routing, relationships	Common parts: PL-300, PU-510, PU-520, PU-530, PK-600; T10/T12 share PL-200-U and PU-500-U	CSA Table 2	CURRENT	Shared parts can use shared supermarket locations; size-specific parts need clear identification	None	Affects kitting layout and segregation	Firm
E-26	Product, routing, relationships	Physical size, weight and container of each part (totes, trays, cartons, pallets)	Not given in the case	VERIFY	Only the container type is known for shafts (partitioned trays) and WIP (totes)	Request unit-load data in the next data release	Needed for handling equipment and storage space	Open
E-27	Product, routing, relationships	Finished shafts are stored in partitioned trays	CSA Table 3	CURRENT	Shaft tips need protected handling between finishing, inspection and kitting	None	Affects handling method and storage	Preliminary
E-28	Product, routing, relationships	Five current routes: molded, machined, purchased, final device, packaging materials	CSA Table 3, Fig. 4	CURRENT	Basis for the activity list and flow relationships	None	A route change alters activities and relationships	Firm
E-29	Product, routing, relationships	Location of degate/trim and visual inspection of molded parts	CSA Table 3; not shown in Fig. 3	ASSUMED	Assumed at press side inside molding; dimensional checks at Q-1/Q-2	Confirm with the data release	Changes the molding–lab relationship	Preliminary
E-30	Product, routing, relationships	The 2 component dimensional-inspection stations serve both molded parts and shafts	CSA Table 4 (one resource type for components)	ASSUMED	Q-1 and Q-2 are shared by streams A and B	Confirm inspection assignment and sampling plan	Separate stations would change lab load and location	Preliminary
E-31	Product, routing, relationships	Off-line rework is co-located with functional test (RM-1 in the test room)	CSA Fig. 3, §2j	CURRENT	Test and rework have a strong flow relationship	None	Separation would add a return flow	Firm

Figure 10. Evidence Assumptions and Verification Register (E-24-E-31)

E-32	Product, routing, relationships	Packaging materials (pouch, labels) form a separate stream that meets the device only at packaging	CSA Fig. 4, Table 3	CURRENT	Packaging needs a link to purchased storage, not to kitting	Confirm staging location (E-21)	Affects storage–packaging relationship	Preliminary
E-33	Product, routing, relationships	Forklifts are excluded from the controlled assembly area	CSA §2a	CURRENT	Assembly is supplied by carts from kitting; forklift-served areas need separation from assembly	Confirm room requirements (E-11)	Shapes closeness and separation ratings	Preliminary
E-34	Product, routing, relationships	Molding, machining and maintenance should be separated from controlled assembly (oil, chips, particulates)	Team assumption from process nature; not stated in the case	ASSUMED	Treated as a separation relationship	Confirm controlled-room requirements (E-11)	Changes relationship ratings	Preliminary
E-35	Quantity, timing, services, activity model	P–Q: T5 is the largest product in every year; T12 grows 56% from Y1 to Y5 and T10 grows 22%	CSA Table 1	CURRENT	Low variety and high volume across the family; the mix shifts toward T12	None	Changes which product drives each resource	Firm
E-36	Quantity, timing, services, activity model	Required pace: 32.8 s/unit (Y0), 23.6 s/unit (Y1), 19.4 s/unit (Y5)	Derived from CSA Tables 1 and 5	CURRENT	Year 1 needs 39% more output per day than Year 0	None	Changes station requirements	Firm
E-37	Quantity, timing, services, activity model	Test yield after rework: T5 99.25%, T10 98.70%, T12 98.10%	Derived from CSA Table 9	CURRENT	Upstream quantities must exceed good demand by 0.8% to 1.9%	None	Changes component and assembly quantities	Firm
E-38	Quantity, timing, services, activity model	Mold changeover and setup times between variants	Not given in the case	VERIFY	Not included in the Appendix A check	Request in the data release	Reduces available press time	Open
E-39	Quantity, timing, services, activity model	Production lot sizes and supermarket replenishment rules	Not given in the case	VERIFY	Needed to size WIP and supermarket	Request or define assumptions	Changes storage and handling frequency	Open
E-40	Quantity, timing, services, activity model	Demand for Years 2 to 4	Not given in the case	ASSUMED	Planning uses Year 1 (requirement) and Year 5 (horizon) only	Confirm with the instructor	Intermediate phasing could change	Preliminary

Figure 11. Evidence Assumptions and Verification Register (E-32-E-40)

E-41	Quantity, timing, services, activity model	Supporting services present in Year 0: quality lab, maintenance/tooling, utilities, offices, lockers/PPE, break room, restrooms, first aid	CSA Fig. 3	CURRENT	Services exist for one shift	None	Base list for the activity model	Firm
E-42	Quantity, timing, services, activity model	Gowning or entry requirements for the controlled room	Not given; implied by "controlled assembly room"	VERIFY	The locker/PPE room is assumed to support room entry	Confirm with E-11	Adds a gowning relationship to assembly	Open
E-43	Quantity, timing, services, activity model	Staffing by area and shift	Not given in the case	VERIFY	Needed to size personnel services	Request in the data release	break and restroom needs	Open
E-44	Quantity, timing, services, activity model	handling (molded and shaft scrap, device scrap)	CSA §2g–§2j (scrap exists); no area shown	VERIFY	A scrap accumulation point is needed	Confirm disposal method and location	Adds a supporting activity	Open
E-45	Quantity, timing, services, activity model	and lathes (cooling water, compressed air, power)	CSA Fig. 3 (Utilities/Mechanical, Feed); not quantified	VERIFY	Utilities room serves molding	Request utility requirements (with E-10)	Constrains where molding can be located	Open

Figure 12. Evidence Assumptions and Verification Register (E-41-E-45)

XI. Current engineering assessment

Based on what we know so far, the current facility may not be able to handle Year 1 at three resources: cannula subassembly, the Swiss CNC lathes and, to a lesser extent, obturator subassembly. On top of that, the device must cross a shared aisle twice on its way through the plant. At this stage, though, we cannot say how much equipment or space will be needed, or which arrangement would work best.

What we can say with confidence is that the case clearly defines demand, product mix and the operating basis for Years 0, 1 and 5 (E-03, E-04). The three products share one architecture: PL-300, PU-510, PU-520, PU-530 and PK-600 are common to all of them, and T10 and T12 share the universal seal housing and instrument seal (CSA, Table 2). The routes for the five material streams, including the return loops caused by rework, are documented (E-15, E-16), and we know which parts of the building are fixed (E-01).

Looking at the evidence, we see four main concerns and one opportunity. The first concern is capacity. A rough load check puts cannula subassembly at about 97% in Year 1 and 120% in Year 5, and the Swiss CNC lathes at about 95% and 119%, which leaves little or no margin (CSA, Tables 5, 7 and 8; E-04, E-05). The second is the device route. Because it crosses the aisle twice and backtracks between assembly, testing, and packaging, the 39% increase in volume in Year 1 means even more of those trips (*Figure 2*, observation 3; E-09, E-11).

The third concern is the main aisle itself. It is only 10 ft wide; it carries all five streams, and it mixes forklifts, pallet jacks, and carts right next to the door of the controlled room (*Figure 2*, observation 2; E-09). The fourth concern is an inspection. Inspection for three streams happens in the northwest corner, so shafts that need rework must repeat the trip between finishing and the lab (*Figure 2,* observation 4; E-12, E-15). On the positive side, there is space that could be put to better use: the 1,300 ft² training/legacy area, and the 8,800 ft² reserve right next to the north side of the building (E-07, E-19). It also helps that T12 only adds three unique parts (its cannula, shaft, and label), so the new variety at kitting stays small (E-14).

There are also several things we cannot conclude yet. We do not know the final number of machines and stations, since that depends on mold changeovers, allowances, and staffing, which are outside the scope of this report. We also cannot confirm whether molding has enough capacity. The rough load is about 66% in Year 1 and 82% in Year 5, but mold-to-press compatibility and changeover times are still missing (E-06). Space and storage needs depend on shipment frequency and unit loads, which the case does not provide (E-09, E-20). Finally, which closeness relationships matter most depend on Section IX and on the requirements for the controlled room (E-11).

Because of this, the items we most need to verify are mold-to-press compatibility, since it decides whether IM-2 becomes a bottleneck once T12 is added (E-06). We also need the actual building dimensions from the DWG, which every area and distance depends on (E-07), and trip frequencies and unit loads, which drive flow intensity (E-09). The controlled-room requirements will tell us which activities need to be close together or kept apart (E-11). Staffing and allowances at the manual stations will confirm or rule out the first concern (E-05).

For the next stage, we will need a proper capacity calculation for each resource that includes changeovers and allowances, building on Section VI. We will also need a from-to chart showing flow intensity between activities, based on Sections V and VIII, and an activity relationship chart and diagram using the reasons in Section IX. With those in place, we can estimate space and storage requirements for each activity in Years 1 and 5.

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